Suggested Solutions:

No.	Solution
1 (a)	From Fig. 1.2, unstretched length of spring = 0.40 m When stretched to length of 0.70 m by a mass of 0.450 kg, extension in spring = 0.30 m Elastic potential energy stored $=\frac{1}{2}kx^2$ $=\frac{1}{2}Fx$ $=\frac{1}{2}(0.450)(9.81)(0.300)$ $=0.6622$ ≈ 0.662 J
	Note: Candidates should read the question carefully and appreciate that the quantities plotted on the graph were not force and extension.
(b)(i)	Between l = 40.0 cm and equilibrium position at l = 60.0 cm, gravitational potential energy decreases while kinetic energy and elastic potential energy increase. Between equilibrium position at l = 60.0 cm and lowest point at l = 80.0 cm, gravitational potential energy and kinetic energy decrease while elastic potential energy continues to increase.
	Note: There seemed to be confusion between zero and minimum gravitational potential energy at the lowest point.
(ii)	By conservation of energy, Loss in gravitational potential energy = Gain in kinetic energy + Gain in elastic potential energy $mg\Delta h=\frac{1}{2}mv^2+0.6622$ $\frac{1}{2}(0.300)v^2=(0.300)(9.81)(0.300)-0.6622$ $v=1.213$ $\approx 1.21\text{ m s}^{-1}$
	Note: The majority of candidates were unable to produce a correct link between the terms representing the energy changes as the spring extends.
(iii)	By conservation of energy, Loss in gravitational potential energy = Gain in elastic potential energy (because mass was released from rest, i.e. no initial kinetic energy)

To find spring constant,

$$mg = kx$$

$$k = \frac{mg}{x}$$

$$= \frac{(0.450)(9.81)}{0.300}$$

$$= 14.715 \text{ N m}^{-1}$$

$$mgx = \frac{1}{2}kx^2$$

$$x = \frac{2mg}{k}$$

$$= \frac{2(0.300)(9.81)}{14.715}$$

$$= 0.400 \text{ m}$$

Note: A large number of candidates attempted to use equations for constant acceleration.